

Your grade: 100%

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To pass you need at least 80%. We keep your highest score.

Next item →

1. Suppose that a continuous-time system has $A = -0.1$, $B_w = 1$, $C = 1$, $S_v = 0.1$, and $S_w = 0.1$. Solve the continuous-time algebraic Riccati equation (ARE) for the steady-state solution to $\Sigma_{\hat{x}}$. Enter your answer with two digits to the right of the decimal point. 1 / 1 point

Hint: The ARE becomes a quadratic equation for this scalar system, which you can solve by hand. Also note that covariance matrices must be positive definite.

0.09

✔

Correct

Yes. That is correct.

2. Suppose that a continuous-time system has steady-state solution $\Sigma_{\hat{x}} = 1$, $C = 1$, and $S_v = 0.1$. What is the steady-state Kalman gain L ? Enter your answer with one digit to the right of the decimal point. 1 / 1 point

10

✔

Correct

Yes. That is correct.

3. Suppose that a continuous-time system has $A = -0.1$, $C = 1$, and steady-state Kalman gain $L = 1$. What is the pole location s of the steady-state Kalman filter? Enter your answer with one digit to the right of the decimal point. 1 / 1 point

-1.1

✔

Correct

Yes. That is correct.